

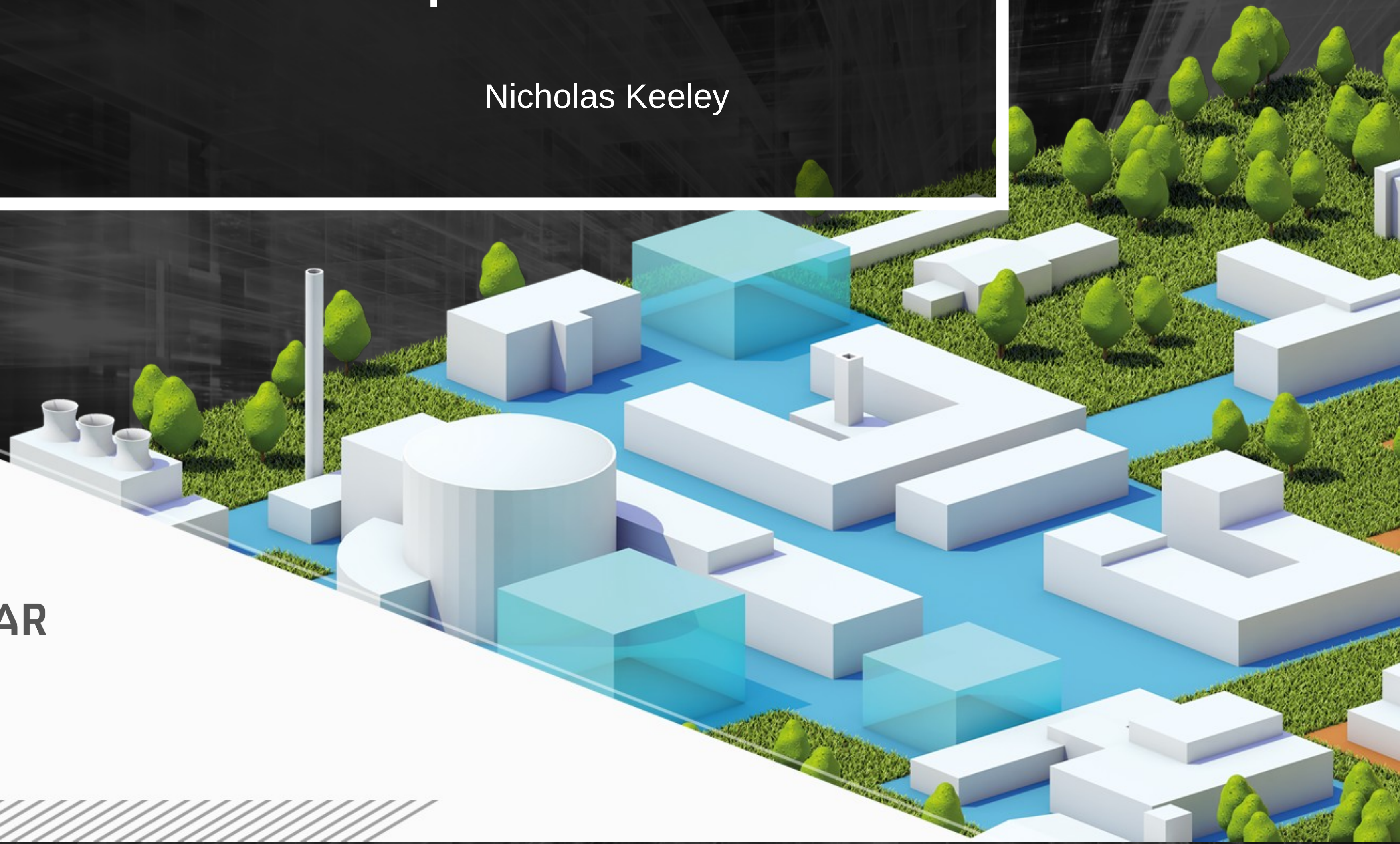


Direct reactions; some practical hints. Part 3: breakup

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Breakup represents one of the most challenging problems in direct reaction work, from both the experimental and theoretical points of view. In this lecture we shall examine the techniques used to analyse breakup data and model the influence of coupling to breakup degrees of freedom on the elastic scattering.

Before we begin we should define exactly what we mean by “breakup” within the context of this lecture. As we shall see, this is by no means as obvious as it might seem ...

If a nucleus is loosely bound it may be easily excited above the threshold for particle emission such that it breaks into two (or more) particles or clusters.

This is an inelastic scattering process to a final “state” which is unbound. This may be an actual resonant state of definite spin-parity and more or less narrow width or the non-resonant continuum.

Breakup via a resonance is sometimes known as “sequential breakup” and to the continuum as “direct breakup”.

If the target is not excited (i.e. it remains a spectator) then both sequential and direct breakup processes are sometimes collectively referred to as “elastic breakup”. This is the most common occurrence; if the target is also excited (equivalent to mutual excitation) we have “inelastic breakup”.

Finally, under the usual definition of breakup none of the fragments is captured by the target. In practice, this is only an issue for direct breakup since most resonant states are sufficiently long-lived they do not decay until they are beyond the nuclear field of the target.

In what follows “breakup” will denote inelastic excitation to either a resonant state or the continuum where the target remains in its ground state and none of the resulting projectile fragments is subsequently captured.

Having defined what we mean by breakup, how do we go about describing it? Since it is basically an inelastic scattering process would the usual CC formalism work?

It is a reasonable approximation for resonances (although it cannot deal with their finite width) but the continuum presents an infinite number of “states”. How to proceed?

We use an extension of the CC formalism called Coupled Discretised Continuum Channels (CDCC) where the continuum is replaced by a finite number of bins in momentum (k) and relative angular momentum (ℓ) space and averaged wave functions calculated over the finite widths of these bins.

Each bin then becomes a “channel” in a standard CC calculation. Exactly how many bins are necessary to obtain a converged result will depend on both the system and the incident energy.

The easiest way to show how this is done will be to take a concrete example. Let us suppose that we wish to calculate the breakup cross section for 34-MeV ${}^6\text{Li} + {}^{58}\text{Ni}$ and at the same time investigate the coupling effect of breakup on the elastic scattering.

${}^6\text{Li}$ has a low threshold (1.4743 MeV) against breakup into an α particle and a deuteron. The relevant breakup mode (indeed the only one observed for this system at the energies we are interested in) is ${}^6\text{Li} \rightarrow \alpha + d$.

We therefore assume an $\alpha + d$ cluster structure for ${}^6\text{Li}$.

We will go into more detail later but for the present we examine the implications of this model for the continuum binning scheme.

We noted earlier that the continuum is divided into “bins” in two ways: we cut it up into chunks as a function of k , the relative momentum of the deuteron with respect to the α particle (the “core”) which are then further split into different relative angular momentum values, ℓ .

This sounds complicated but our example should make it clear.

Ignoring the split in ℓ for the moment we concentrate on the k binning. Firstly, we define k :

$$k(\text{fm}^{-1}) = 0.2187[m(\text{u})E(\text{MeV})]^{1/2}$$

Where m is the reduced mass of the $\alpha + d$ system (8/4) and E their relative energy (this corresponds to the “excitation energy” of the unbound ${}^6\text{Li}$ above the $\alpha + d$ threshold).

k is a continuous variable which we now divide up into constant width chunks from $k = 0$ up to $k = k_{\text{max}}$. We set k_{max} by trial and error; stop when adding a further bin makes a negligible change to the observables.

Why k rather than E (especially since in an actual calculation with FRESKO we specify the bins as functions of E ...)?

It has been found that equally spaced k bins are adequate for convergence of the binning scheme and thus reduce the number of bins required for a given $k_{\max}$ compared to equally spaced E bins (recall $k \sim E^{1/2}$).

It is also found that a bin width of $\Delta k = 0.1 \text{ fm}^{-1}$ gives good convergence of both calculated elastic scattering and breakup, so we will assume $\Delta k = 0.1 \text{ fm}^{-1}$ in what follows.

k		E
0.5	3.2146	3.9202
0.4	1.9601	2.5089
0.3	1.0193	1.4113
0.2	0.3920	0.6272
0.1	0.0784	0.1568
0.0		0.0000

The scheme on the left shows the basic bin structure up to $k_{\max} = 0.5 \text{ fm}^{-1}$. This takes care of the discretisation of the continuum in k -space (subject to convergence tests for the final value of $k_{\max}$).

How do we split this up for the different ℓ values?

ℓ is a non-continuous variable (quantised) and here can take any integer value from 0 to ∞ , although we truncate at $\ell_{\max}$ in a similar way to k (test for convergence). In practice, $\ell_{\max} = 3$ or 4 is usually sufficient.

Thus we may simply duplicate the previous scheme $\ell_{\max}$ times and we are done? Not quite ...

Each of our bins must have a definite spin-parity if we are to include it in a CC scheme within the CDCC formalism. ℓ gives us a definite parity, but we need to work out a total spin j .

The ground state of the α particle is 0^+ and of the deuteron 1^+ . Therefore, in an ℓsj coupling scheme, for a given $\alpha + d$ relative angular momentum of ℓ , since $s=1$ here (the deuteron spin) j can take any integer value such that:

$$\ell-1 \leq j \leq \ell+1$$

Thus for $\ell=0$, $j^\pi=1^+$

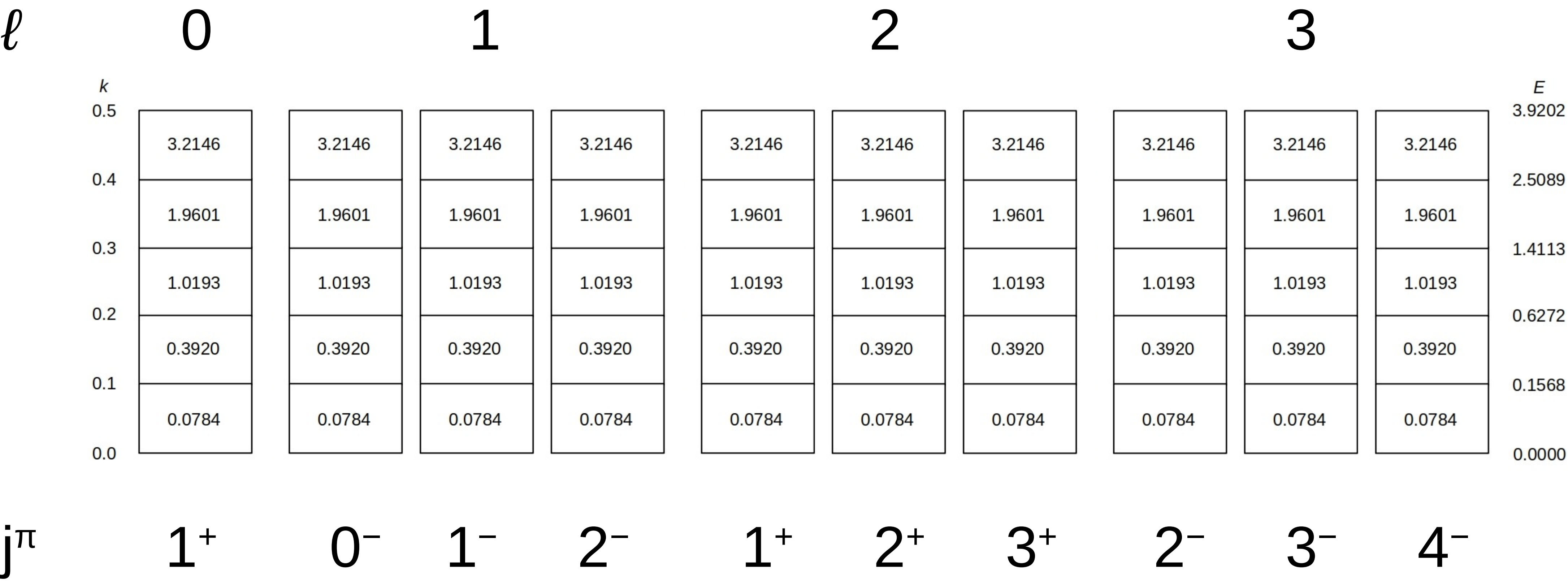
$$\ell=1, j^\pi=2^-, 1^-, 0^-$$

$$\ell=2, j^\pi=3^+, 2^+, 1^+$$

$$\ell=3, j^\pi=4^-, 3^-, 2^- \text{ etc.}$$

so that for $\ell_{\max} = 3$ we need to expand our binning scheme x10! Note that some j^π combinations can result from more than one value of ℓ .

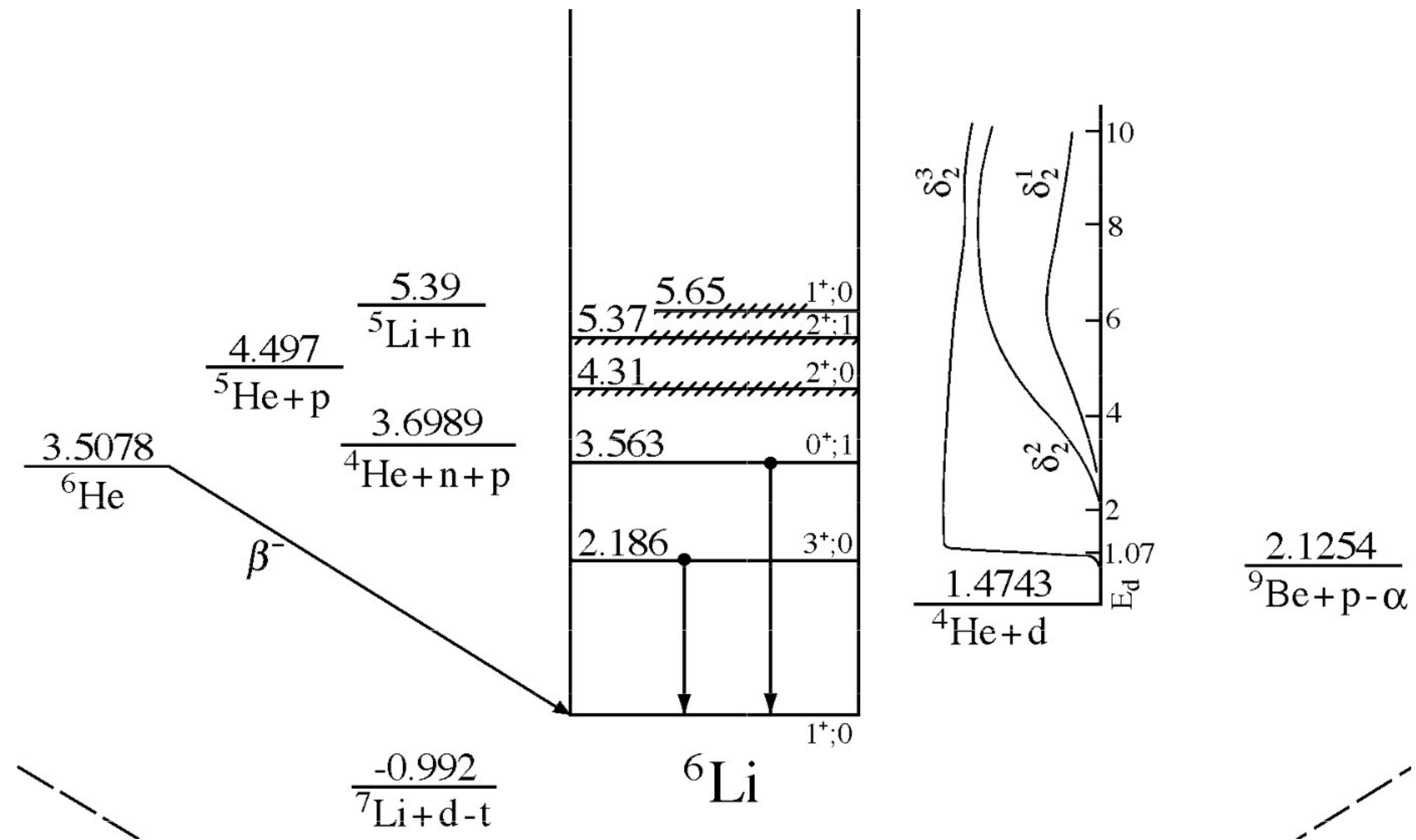
This gives us a scheme something like this:



This would be our final binning scheme for a nucleus with no resonant states embedded in the continuum.

However, if we look at the level scheme for ${}^6\text{Li}$ we see the following:

There are three narrow resonances that concern us: the 2.186 MeV 3^+ , the 4.31 MeV 2^+ and the 5.65 MeV 1^+ . These all have an $\alpha + d$ cluster character with $\ell = 2$. Their widths are 0.024 MeV, 1.30 MeV and 1.5 MeV respectively (from the TUNL database)



We must modify our binning scheme to take account of them.

Resonances require special handling, so we need a separate bin for each of them. The bin must be wide enough to accommodate the whole strength but not too wide. We adopt the following energy (E) widths: 3^+ 0.1 MeV, 2^+ 2.0 MeV and 1^+ 3.0 MeV.

This means that we must modify our $\ell = 2$ binning scheme to take this into account. To avoid double counting we must leave gaps in the non-resonant continuum where the resonant bins occur ...

As an illustration I show the revised binning scheme for the $j^\pi = 1^+, \ell = 2$ continuum including the resonance (coloured red).

This case does not perturb the scheme significantly and we soon get back to the “standard” sequence of bins.

The schemes for $\ell = 2, j^\pi = 2^+$ and $j^\pi = 3^+$ are modified in a similar way.

k		E
0.7	6.7049	7.6835
0.6043	4.2263	5.7263
0.4169	2.0688	2.7263
0.3	1.0193	1.4113
0.2	0.3920	0.6272
0.1	0.0784	0.1568
0.0		0.0000

Now that we have defined our bins, each of which corresponds to an “excited state” of definite spin-parity in a standard CC scheme we need to calculate the couplings between them.

I will describe the procedure used in FRESCO since this is the only generally available code which can handle CDCC.

We use the cluster folding model to calculate the potentials (diagonal and coupling) we need. It is also sometimes known as Watanabe folding, first applied to deuteron scattering.

We do not have time to go into the details of the underlying theory, but to calculate the potentials we assume the $\alpha + d$ cluster model of ${}^6\text{Li}$ and fold $\alpha + {}^{58}\text{Ni}$ and $d + {}^{58}\text{Ni}$ OMPs with the appropriate $\alpha + d$ internal wave functions of the ${}^6\text{Li}$ “states” concerned (i.e. the ground state and the various continuum bins).

To begin, how do we calculate these wave functions? For the ${}^6\text{Li}$ ground state this is relatively easy. It is calculated in exactly the same way as for a bound state in a transfer reaction. We just need to define N, L, J and choose a binding potential.

The ${}^6\text{Li}$ ground state is 1^+ , so the ℓ of the deuteron relative to the α core is 0 or 2. However, the $\ell = 2$ component is so small that we may safely neglect it for our purposes and assume a pure $\ell = 0$ configuration (“spectroscopic factor” of 1.0).

Since we know the $\alpha + d$ binding energy (1.4743 MeV) we just need a binding potential. We take an ordinary Woods-Saxon with parameters adjusted to yield a wave function which reproduces known properties of ${}^6\text{Li}$ (rms charge radius etc.): $R = 1.90$ fm, $a_0 = 0.65$ fm, depth adjusted to give correct binding energy.

For the continuum things are a little more complicated. The binding energy is now negative (unbound). If we just calculate a single energy eigenstate (as we would for a bound state) it will not converge to produce a square-integrable wave function.

This problem is solved by taking an averaged wave function over a range of energies (the width of the bin). With some mathematical tricks square integrable wave functions result with normalisations very close to 1.0.

To calculate these wave functions we use the same binding potential as for the ground state. For most continuum bins we hold the potential depth fixed at the same value as for the ground state.

Resonant bins require special handling. Most importantly, we must arrange things such that there is indeed a resonant wave function at the required energy.

There are several definitions of a resonance, the one we use is that the resonance occurs at the energy E where the phase shift angle $\delta = 90^\circ$. FRESCO will print δ as a function of E , and we adjust the depth of the binding potential (for the resonant bin wave function) until $\delta = 90^\circ$ at E_{res} , the resonant energy (N.B. $E_{\text{res}} = E_{\text{ex}} - 1.4737 \text{ MeV}$ for ${}^6\text{Li}$).

For a consistent result, this binding potential depth should also be used for the non-resonant continuum bins of the same spin-parity as the resonance.

One final point concerning these averaged wave functions. Due to the way they are normalised, they must be calculated out to quite large radii to obtain normalisations close to unity (as required). These can be of the order of 100 fm or more.

Finally, these wave functions are used together with the OMPs for $\alpha + {}^{58}\text{Ni}$ and $d + {}^{58}\text{Ni}$ for 4/6 and 2/6 of the incident ${}^6\text{Li}$ energy to calculate the required diagonal and coupling potentials for all “channels” (the various bins). The “excitation energy” of each bin is simply the mid-point energy (of the bin limits) + 1.4737 MeV (for resonances we arrange things such that this is the observed resonance energy).

These are then used in a standard CC calculation.

In an actual calculation, FRESKO does all the hard work for us, calculating all the wave functions and diagonal and coupling potentials. You will find a comprehensive description of the underlying theory in Prog. Theor. Phys. Suppl. **89**, 1 (1986).

FRESKO will calculate all couplings, including those between continuum bins (these are found to be important) up to a specified multipolarity, λ . Usually, $\lambda = 4$ is sufficient (although, depending on the system, if you include $\ell > 4$ in your continuum scheme you may need to increase λ).

Let us now see how this works in practice. We return to our specific example of ${}^6\text{Li} + {}^{58}\text{Ni}$ at an incident energy of 34 MeV.

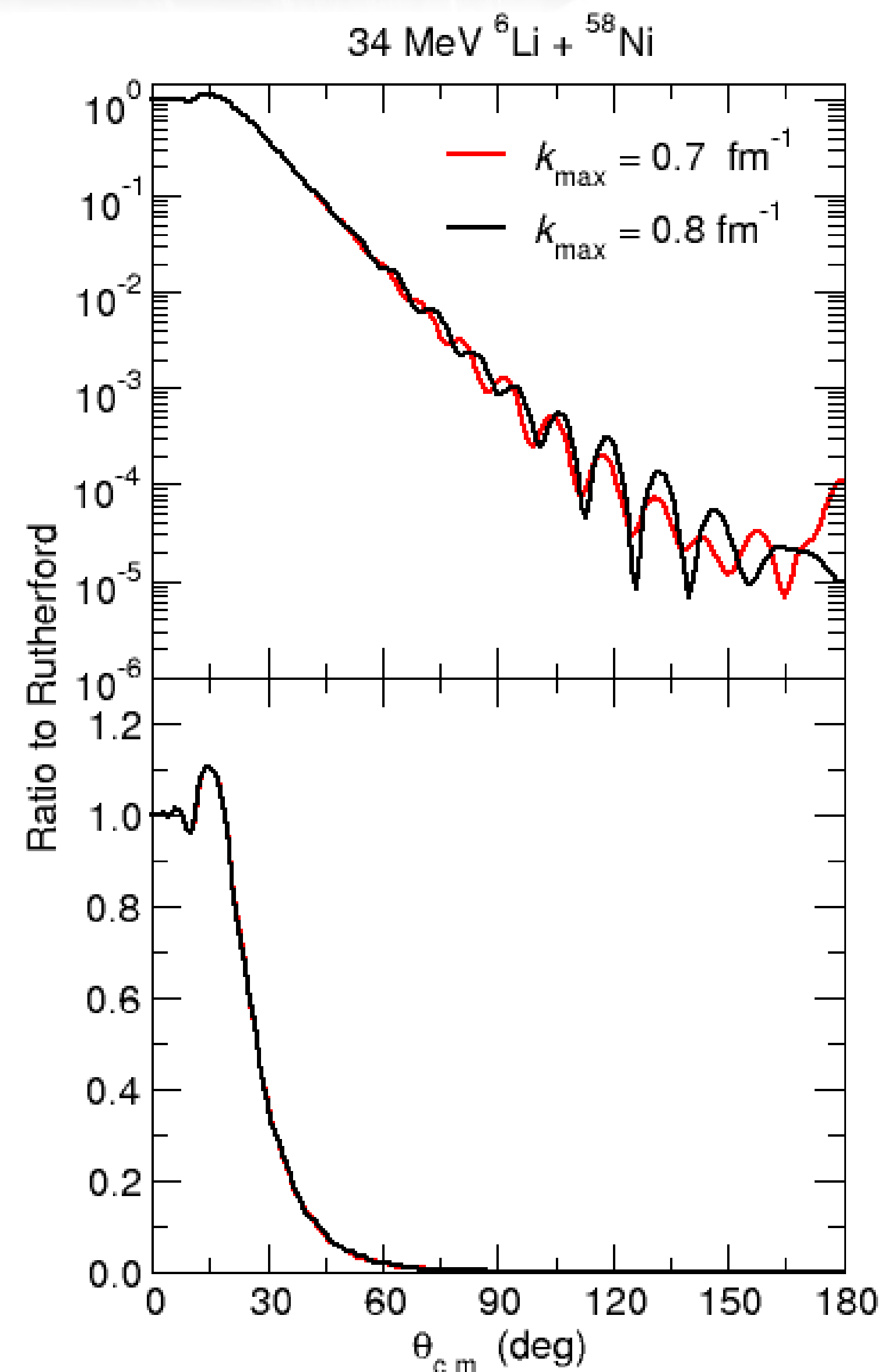
Elastic scattering data are available from K. D. Veal et al., Phys. Rev. C **60**, 064003 (1999). We need to supply $\alpha + {}^{58}\text{Ni}$ and $d + {}^{58}\text{Ni}$ OMPs at 22.7 and 11.3 MeV respectively. We could use global parameter sets, but there are empirical OMPs for these systems at 23 and 11.8 MeV, see the compilation of Perey and Perey, Atomic Data and Nuclear Data Tables **17**, 1 (1976).

Our first step is to check convergence of the elastic scattering and breakup as a function of k_{max} ($\Delta k = 0.1 \text{ fm}^{-1}$ in all cases). For this test we only include the $\ell = 0$ non-resonant continuum.

We first try $k_{\text{max}} = 0.7 \text{ fm}^{-1}$ and compare the result with that for $k_{\text{max}} = 0.8 \text{ fm}^{-1}$.

Elastic scattering is already rather well converged (depending on how precise one wishes to be).

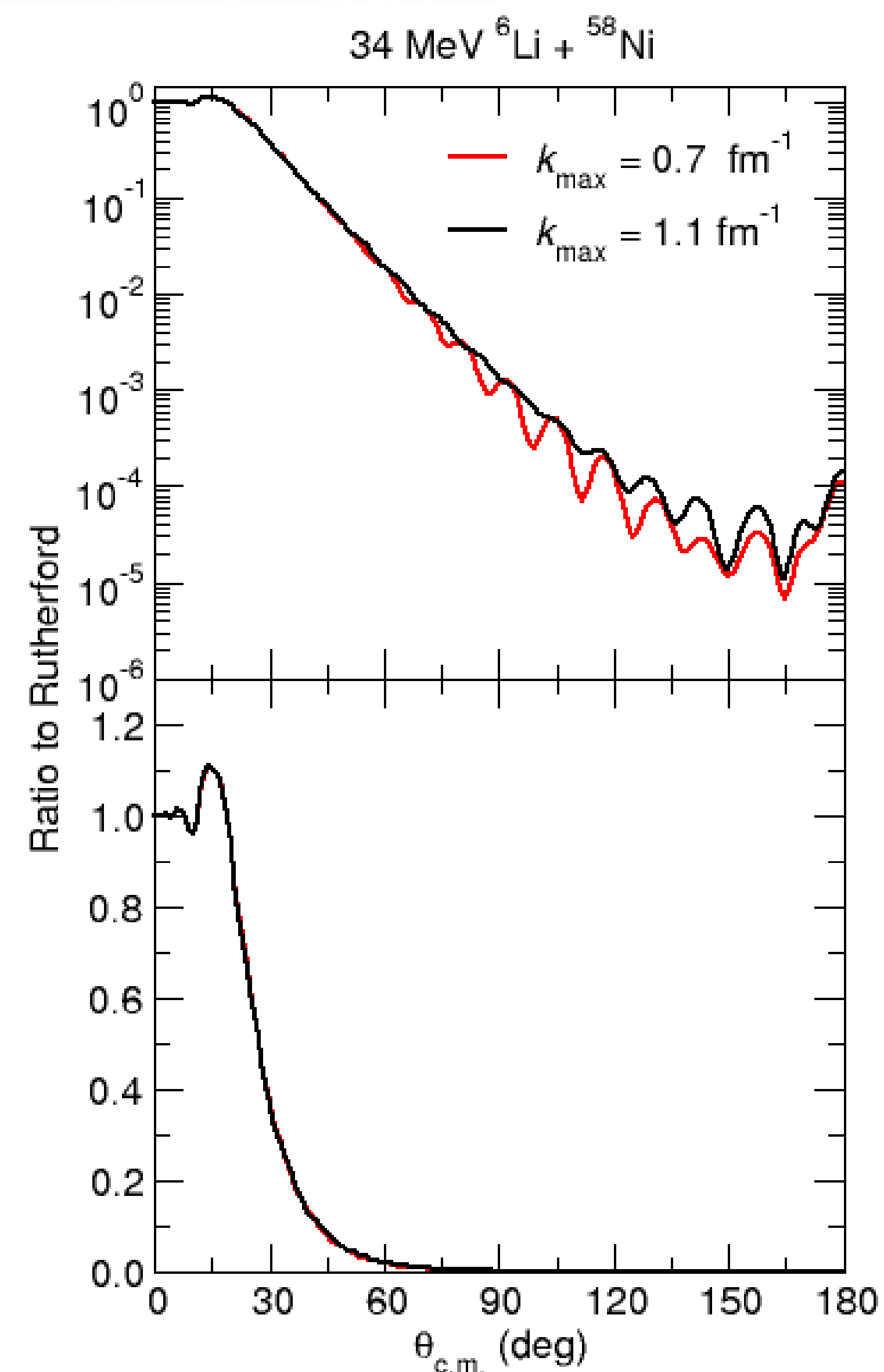
Breakup is less well converged: for $k_{\max} = 0.7 \text{ fm}^{-1}$, $\sigma_{\text{bu}} = 37.4 \text{ mb}$. For $k_{\max} = 0.8 \text{ fm}^{-1}$, $\sigma_{\text{bu}} = 33.8 \text{ mb}$. This is a significant difference, so we will need to check the effect of increasing $k_{\max}$ still further. Note that, contrary to what might naively be expected, increasing $k_{\max}$ does not necessarily increase σ_{bu} .



Continuing the exercise, we find the following:

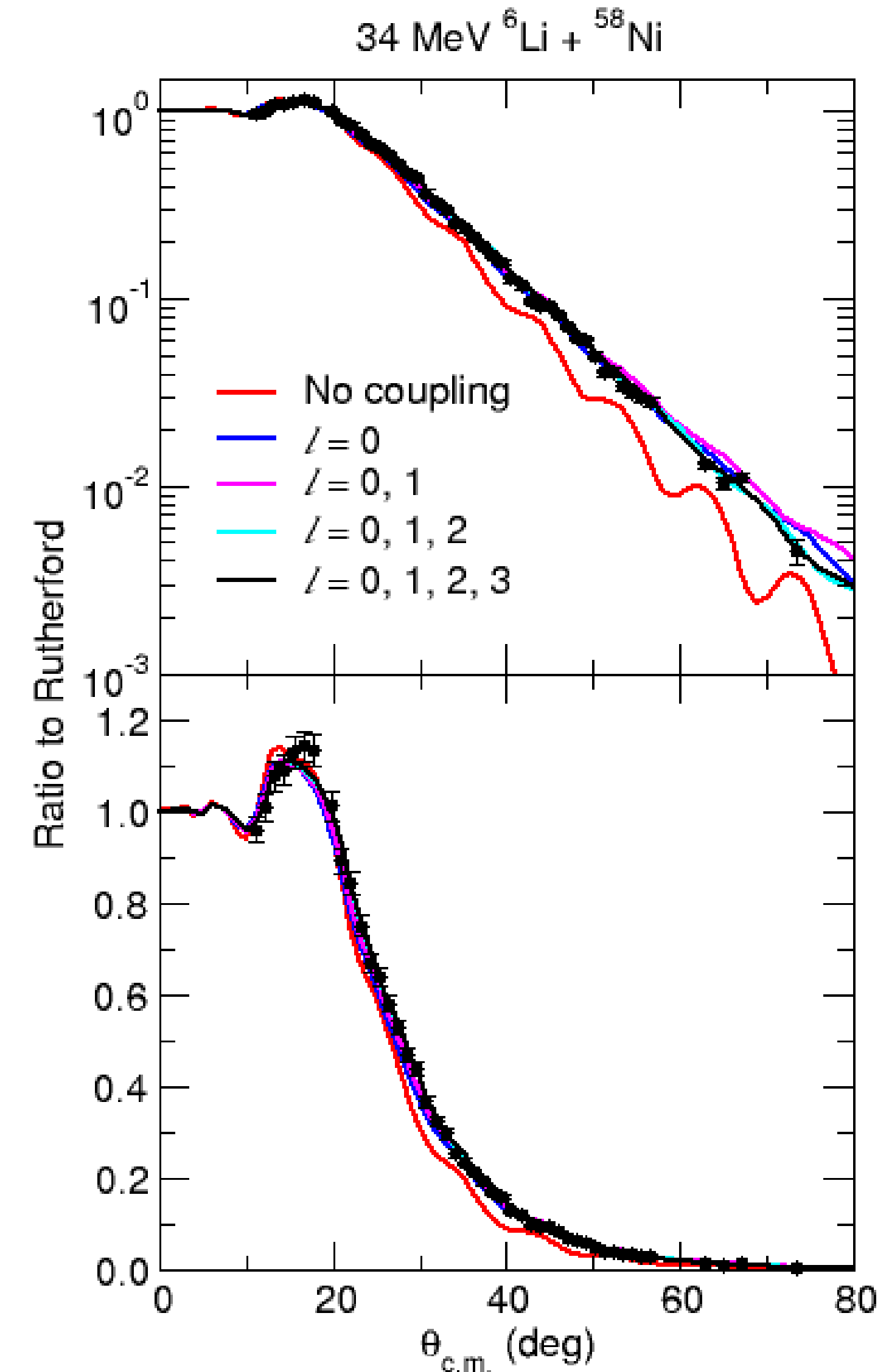
$k_{\max}$ (fm ⁻¹)	σ_{bu} (mb)
0.9	31.6
1.0	30.7
1.1	30.4
1.2	30.3

Effective convergence at $k_{\max} = 1.1$ fm⁻¹ (also for elastic scattering).



We now check the convergence as we increase ℓ , keeping $k_{\max}$ fixed at 1.1 fm^{-1} . For the sake of comparison, we also show the no coupling result.

In this case the breakup coupling effect on the elastic scattering is modest, and simply including the $\ell = 0$ continuum already gives a good description of the elastic scattering data. As far as the elastic scattering is concerned, increasing the limits of our ℓ space for the continuum only changes the details.



However, things are very different when we consider the breakup cross section:

ℓ	σ_{bu} (mb)
0	30.4
0, 1	30.8
0, 1, 2	66.1
0, 1, 2, 3	64.4

Note the large increase when we added the $\ell = 2$ continuum; this is due to the three resonant states. Together they contribute 24.6 mb to the total, of which 18.6 mb is due to the 2.18 MeV 3^+ alone.

Thus, where one truncates the continuum binning scheme will depend very much on the system (not only the weakly bound projectile but also the target, since the relative importance of Coulomb breakup can impact the binning scheme) and the incident energy.

If there are resonances embedded in the continuum it will usually be important to include them if an accurate value for the breakup cross section is required. However, omitting them may not affect the elastic scattering significantly.

As a general observation, the calculated elastic scattering usually converges faster than the breakup cross section.

When performing CDCC calculations it is therefore important to check the adequacy of your continuum space, with regard both to $k_{\max}$ and $\ell_{\max}$ (and possibly to some extent the maximum value of λ). Such checks can be time consuming but are a necessary part of good practice in work of this type.

One may also need to experiment with the matching radius rather more than for a conventional CC calculation since this will usually govern the maximum radius for normalising the continuum wave functions. If it is too small it can also impact the result, the breakup cross section in particular.

One final important point. If you wish to compare the calculated angular distributions for breakup (for example to a resonant state) with breakup data, remember that FRESCO actually performs a standard CC calculation (with some extensions ...) so that the breakup angular distributions are simply of ${}^6\text{Li}$ “inelastic scattering”.

They thus “reconstruct” the angular distribution of the excited ${}^6\text{Li}$ just before it broke up into the α particle and the deuteron. Therefore, breakup data measured in a coincidence experiment must be transformed to perform the same “reconstruction” of the single differential ${}^6\text{Li}$ cross section – a non-trivial task as any of you who have done so will know only too well!

In this lecture in particular I have only scratched the surface of a complex subject. However, I hope the few hints I have been able to give during the time available will be useful.

There are many aspects of CDCC that I have not had time to mention; I have tried to pick out the most important.

Thank you for your attention



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